Founding Team Member: Lead Laser-Matter Interaction & Optical Engineer

Company: Cascade Dynamics

Status: Stealth Mode Deep-Tech Startup

Location: Lab-based deep tech hub in Champaign, IL

Compensation: Meaningful Founding Equity Package (with transitioning base salary post funding round)

About the Company

We are a stealth-mode, pre-revenue deep-tech company engineering a new class of physics hardware at the intersection of light, matter, plasma, and charged particle beams. Powered by a foundational patent portfolio, our platform is developing and testing a proprietary plasma-source architecture intended to reduce reliance on hazardous, high-maintenance, and supply-chain-vulnerable legacy technologies across multiple industrial and scientific applications.

Our founding team brings experience in plasma physics and entrepreneurship. We are seeking a hands-on laser-matter interaction and optical systems engineer to own the pulsed-laser, beam-delivery, and optical-diagnostic work required for a maintained-particle, laser-commanded ECR development program. Our initial commercial focus is expanding access to heavy-ion single-event-effects testing. We have completed I-Corps customer discovery and were invited by NSF to submit a Phase I SBIR proposal.

The Role

As a Founding Team Member and Lead Laser-Matter Interaction & Optical Systems Engineer, you will own the path from controlled light delivery at maintained particles to calibrated evidence of material release. You will select and commission the appropriate pulsed-laser system, build the vacuum-compatible optical train, and work hands-on across integrated test campaigns from initial alignment through troubleshooting and repeatability testing.

You will work directly at the boundary between laser-matter interaction and integrated plasma hardware, translating the selected optical regime into a safe, measurable, and serviceable subsystem. Working with the computational, ECR, and controls leads, you will define optical interfaces, timing requirements, diagnostic needs, and configuration-controlled test campaigns..

Core Responsibilities

- Laser-Particle Interaction & Pulsed-Laser Operations: Select, commission, operate, and maintain the pulsed-laser architecture appropriate to the tested interaction. Establish

reproducible operating conditions through measured laser performance and material-response evidence rather than prescribing a pulse regime in advance.
- Calibrated Beam Delivery: Design, align, and characterize the optical path through the vacuum boundary to the maintained-particle interaction region. Measure delivered pulse energy, beam profile, spot and particle overlap, fluence, repetition rate, pointing stability, timing, drift, and uncertainty at the interaction region.
- Vacuum Optical Integration: Engineer windows, feedthroughs, focusing and steering arrangements, debris shields, replaceable components, and cleaning procedures. Characterize transmission stability, deposition, contamination, thermal drift, damage limits, and service intervals.
- Optical Diagnostics & Material-Release Evidence: Implement calibrated imaging, scattering, spectroscopy, photodiodes, or laser-induced fluorescence where justified to measure particle presence, laser-particle overlap, released material, plume evolution, and plasma response. Correlate optical evidence with synchronized ECR and extracted-beam measurements.
- Timing, Safety & Cross-Functional Execution: Define optical trigger, delay, jitter, and acquisition requirements for integrated experiments. Establish compliant enclosures, shutters, beam dumps, access controls, safe states, and procedures; work with the controls lead on synchronized acquisition and interlock implementation, and contribute supportable results to technical milestones, proposals, and patent disclosures.

Required Qualifications
- Educational Background: Ph.D., M.S., or equivalent experimental record in optics, photonics, laser-matter interaction, applied physics, electrical engineering, or a closely related field.
- Hands-On Laser Systems Experience: Demonstrated experience commissioning, aligning, operating, maintaining, and troubleshooting pulsed-laser experiments and complex free-space optical systems.
- Beam Metrology & Experimental Rigor: Experience calibrating pulse energy, spatial profile, spot size, overlap, fluence, repetition rate, pointing, timing, drift, and uncertainty at a named interaction plane.
- Vacuum & Harsh-Environment Optics: Practical command of vacuum-compatible optical hardware, viewport fouling, contamination and debris control, optical damage, thermal effects, serviceability, and laboratory laser-safety practice.
- Mindset: A high-agency, startup-ready experimentalist committed to sustained hands-on laboratory work in a fast-paced, un-siloed "Skunkworks" environment. Performance dictates corporate progress here.
- Security Clearance: Preferred ability to obtain a U.S. Government Security Clearance (U.S. Citizenship typically required for downstream defense prime deployments).

Preferred Extras
- Experience with laser-produced plasmas, laser ion sources, maintained particles, aerosols, powders, droplets, or optical diagnostics in high-vacuum or magnetized-plasma systems.

- Experience with time-resolved imaging, scattering, optical emission spectroscopy, laser-induced fluorescence, synchronized acquisition, contamination-prone optical systems, or ECR and accelerator-source interfaces.

What We Offer

- Direct Equity Ownership: A true co-founding technical stake in a venture tackling a multi-billion-dollar bottleneck with definitive, validating customer pull.
- High Impact Culture: The autonomy to build a technical pipeline from scratch alongside world-class peers, completely free of corporate bureaucracy or traditional academic slow-downs.